

- c) Standard heat of reaction (CO4)
- d) Mole fraction (CO2)
- e) Density of gas Mixture (CO2)
- f) Charle's Law (CO2)

Q.24 Heat of combustion of CH_3OH is - 725 kJ/mol. If 200 gms of CH_3OH is completely combusted to generate steam at 373K from water at 298K. Calculate the quantity of steam generated. (Latent heat of vaporization of water is 2260 KJ/Kg and sensible heat of water is 4.187 KJ/kg.K.) (CO4)

Q.25 A coke containing 90% carbon and 10% non-combustible material by weight calculate (CO4)

- a) The amount of theoretical air required to burn 200 kg of coke completely.
- b) The composition of gases in the product stream if 20% excess air is supplied.

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**3rd Sem / Chemical, Chemical (Pulp & Paper),
Chem Engg. (Spl. Paint Tech.), Chem Engg. (Spl.
Polymer Engg)**

**Subject : Chemical Process Calculations /
Ind. Chem Cal.**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 For Ideal gases, heat capacity relation is (CO2)

- a) $C_p = C_v - R$ b) $C_v = C_p + R$
- c) $C_p - C_v = R$ d) $C_p + C_v = R$

Q.2 One cubic meter (1m^3) is equal to _____ Litres. (CO1)

- a) 10 b) 100
- c) 1000 d) 10000

Q.3 The number of moles of a solute per liter of solution is (CO2)

- a) Molarity b) Molality
- c) Normality d) Gram Equivalent

Q.4 For exothermic reactions heat of formation is (CO4)

- a) Positive b) Negative
- c) Zero d) None of these

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- Q.5 At standard conditions, volume of 1Kg mole of ideal gas is (CO2)
 a) 224m^3 b) 2024m^3
 c) 2240m^3 d) 22.4m^3
- Q.6 Concept of material balance is based upon (CO3)
 a) Conservation of energy
 b) Conservation of momentum
 c) Conservation of volume
 d) Conservation of mass

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define Normality. (CO2)
 Q.8 State Amagat's Law. (CO2)
 Q.9 Define latent heat. (CO4)
 Q.10 Calculate the number of moles present in 54 gm of water (H_2O). (CO2)
 Q.11 Define Combustion. (CO4)
 Q.12 Write the SI unit of Viscosity. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Prove that mole percent is equal to pressure percent. (CO2)

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- Q.14 Calculate the number of kilo moles of H_2SO_4 containing 128 kg of Sulphur. (CO2)
 Q.15 Describe in brief theoretical air and excess air. (CO4)
 Q.16 Explain in brief specific gravity and write its unit. (CO1)
 Q.17 Explain Hess's Law of constant heat summation. (CO4)
 Q.18 List down the basic steps taken for material balance. (CO4)
 Q.19 Derive ideal gas equation. (CO3)
 Q.20 A solid mass of 90 kg containing moisture is dried from 70 percent moisture to 30 percent moisture. Calculate the quantity of water evaporated. (CO3)
 Q.21 Discuss the concept of by pass operation with the help of neat diagram. (CO3)
 Q.22 Calculate the gm weight of NaCl needed to prepare one litre of 2 molar (2M) NaCl Solution. (CO2)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain any four :-
 a) System of units (CO1)
 b) Recycle operation (CO3)

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